

Session: 3032

DFSMSHsm Control Data Set Audit Fundamentals

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Agenda

- **DFSMShsm Control Data Sets**
- **MCDS**
- **BCDS**
- **OCDS**
- **Why Audit the DFSMSHsm Control Data Sets**
- **Auditing the DFSMSHsm Control Data Sets**

Basics

- **DFSMSHsm provides data management facilities for MVS**
 - ✓ Automates space management tasks
 - ✓ Ensures that DASD space is available for successful allocation
 - ✓ Creates backup copies for use in the event of corruption or deletion of data

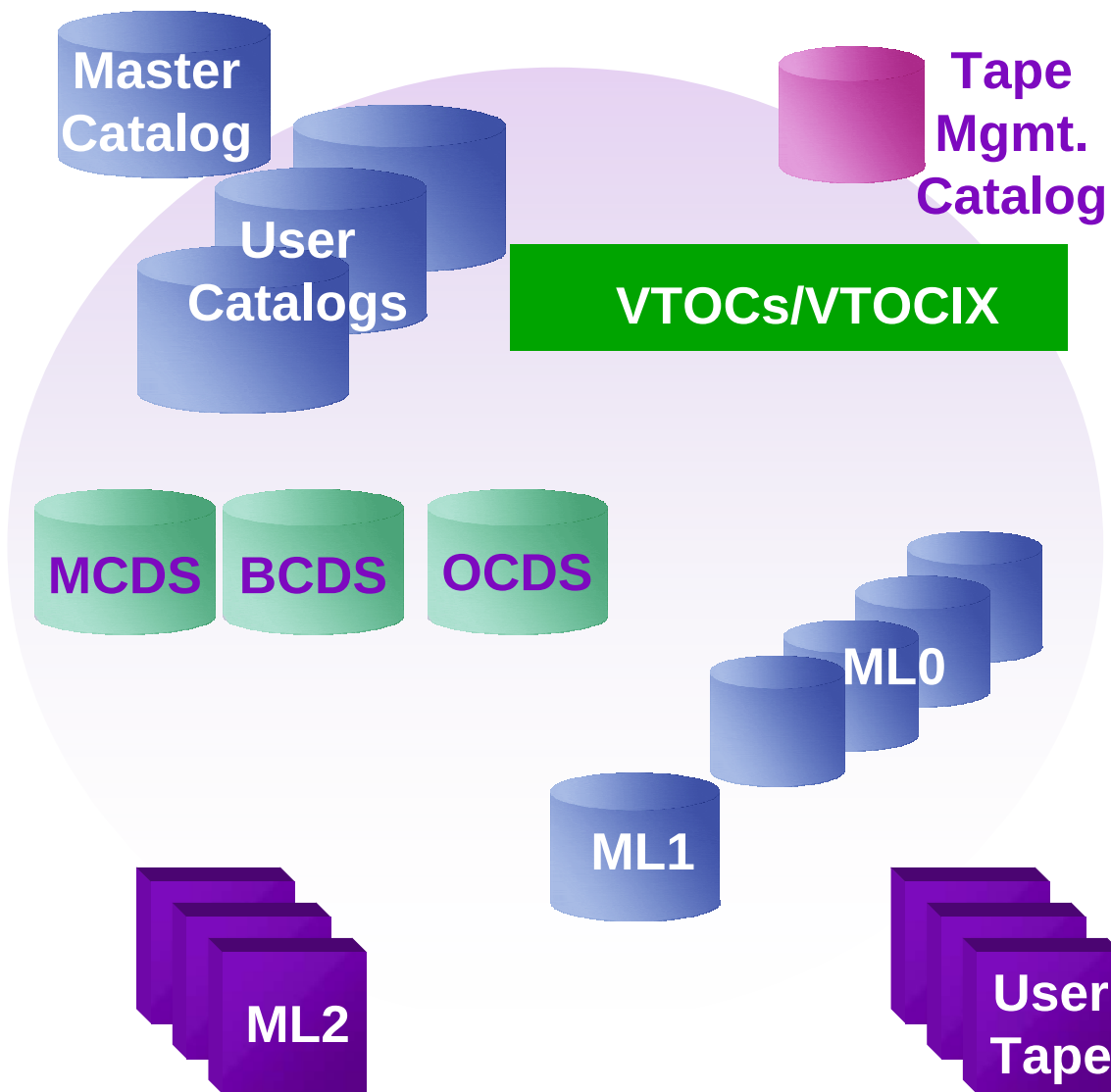
Basics

- Migration Control Data Set
 - ✓ MCDS
- Backup Control Data Set
 - ✓ BCDS
- Offline Control Data Set
 - ✓ OCDS
- Allocate the CDS as VSAM cluster
 - ✓ Multi-clustered VSAM data set
 - Greater than 4GB



DFSMSHsm Control Data Sets

- Control data sets are metadata structures for DFSMSHsm
- Without them, DFSMSHsm cannot function
- When records within the CDS become corrupted, data loss or a disruption in service can occur



Wealth of Information

- **Control Data Sets contain data set information**
 - ✓ **Data set name**
 - ✓ **Last reference, migration, expiration date, and last backup dates**
 - ✓ **Original and migration volumes**
 - ✓ **Original and migration (compressed) space used**
 - ✓ **Number of times data set was migrated**
 - ✓ **VSAM or non-VSAM indicator**
 - ✓ **SMS Class information**

Wealth of Information (continued)

- **Control Data Sets contain statistical information**
 - ✓ Startup and shutdown information
 - ✓ Number of Work Elements processed
 - ✓ Number of ML2 volumes recycled
 - ✓ Requests initiated and failed
 - ✓ Time spent waiting, allocating, processing

Wealth of Information (continued)

- Statistical information can be requested by function
 - ✓ Backup, Delete, Migration, Recall, Recycle, Recover, Spill
- Statistical information can be requested by volume
- Available by using REPORT command
- Statistics can be written to SMF to feed your favorite report writer
- Also, many ISV tools available

Control Data Set Records

- There are as many as 26 different types of records

A	MCA	MCDS alias entry record
B	MCB	BCDS data set record (true name)
C	MCC	BCDS version record
D	MCD	MCDS data set record (true name)
G	DGN	BCDS dump generation record
L	MCL	BCDS backup of a migrated data set
M	MCM	BCDS move backup version record
N	VAC	MCDS JES3 volume activity count
O	MCO	MCDS VSAM association record

Control Data Set Records (continued)

P	MCP	BCDS eligible volume record
Q	ABR	BCDS ABARS record
R	BCR	BCDS backup control record
	BVR	BCDS backup cycle volume record
	DCR	BCDS DUMP control record
S	DSR	MCDS daily statistics record
	L2CR	MCDS ML2 record
	MCR	MCDS management control record
	MHCR	MCDS multiple host processor record
	VSR	MCDS volume statistics record
T	TTC	OCDS Tape Table of Contents (TTOC)

Control Data Set Records (continued)

U	MCU	MCDS user record
V	MCV	MCDS volume record
W	DCL	BCDS dump class record
X	MCT	BCDS backup volume record
Y	DVL	BCDS dump volume record
1	MC1	MCDS ML1 FREESPACE record

Key CDS Records

- Records that are part of any DFSMSHsm audit

MCA MCP

MCB MCV

MCC MCT

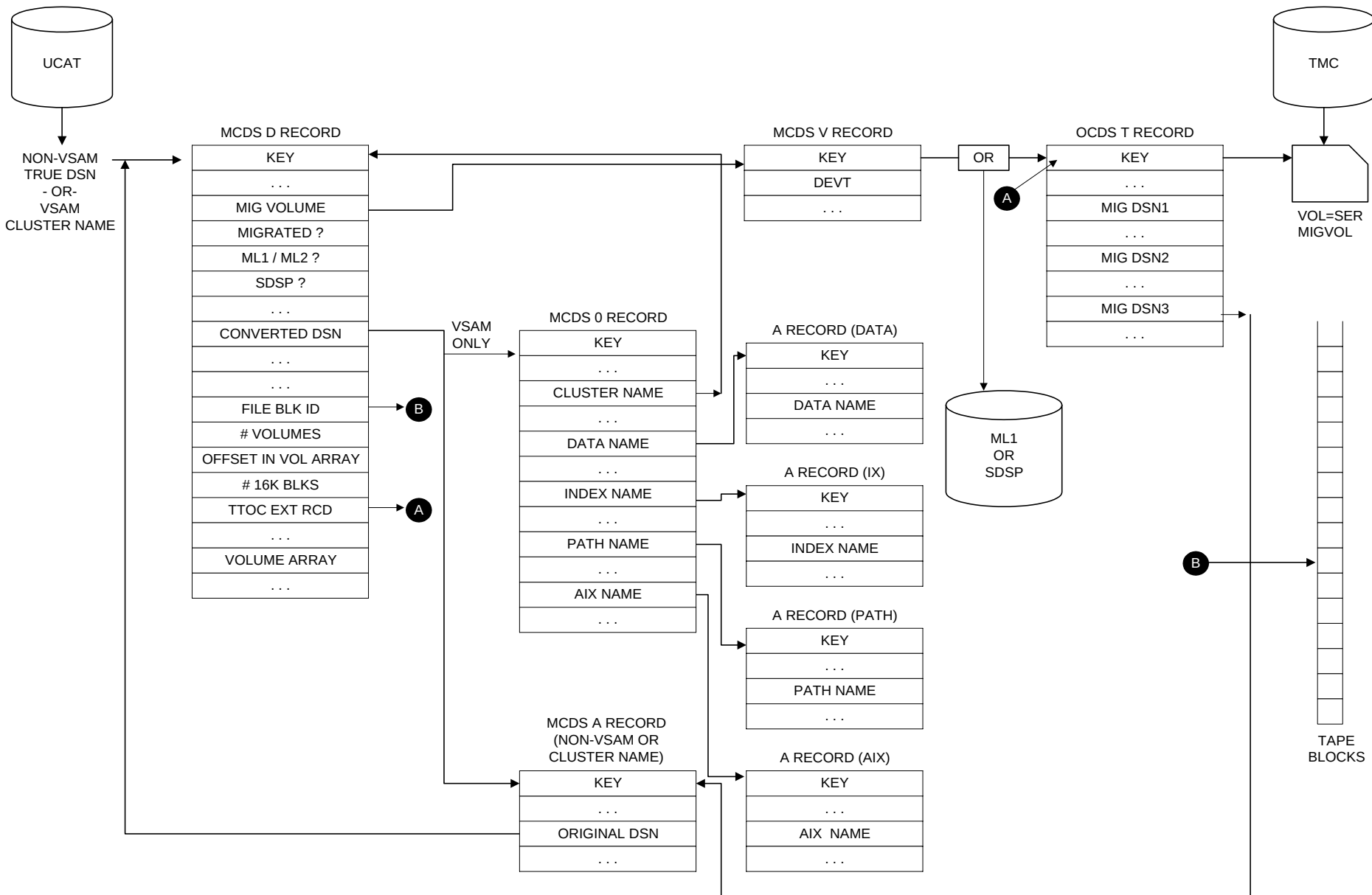
MCD TTC

MCO ABR

IBM Reference Guide:
DFSMSHsm Diagnosis
Reference LY35-0111

CDS Records Are Chained Together

- A record chain is:
 - ✓ A pointer to a physical address within the record that tells DFSMSHsm where to find the next record
- Records can have both forward and backward pointers
 - ✓ MCD record contains pointers to the MCA, MCV, and CDD record
 - ✓ MCA record contains pointers back to the MCD record
- Broken or corrupted pointers can cause recall, recover, and recycle processing to fail



Migration Record Chains

- MCD to MCA – DFSMSHsm generated name
- MCA to MCD – Data set name
- MCD to MCO – DFSMSHsm generated name
- MCO to MCD – VSAM cluster name
- MCO to MCA – VSAM component name
- MCD to MCV – Migration volume
 - ✓ MCV points to either ML1 or ML2 volume
- MCD to TTOC – TTOC Extension record
- MCD to ML2 tape – File Block ID
- TTOC to MCA – DFSMSHsm generated name
- MCD to ICF Catalog – Data set name
- TTOC to Tape Management – Migration volume

Migration Records

■ MCD Record

✓ Migration Control Data Set Record

- User data set name
- VOLSER of migration volume
- User volume (where the data set came from)
- Device type of user volume
- DFSMSHsm migration data set name
- File Block ID
- TTOC extension record

Migration Records (continued)

- MCD Record (continued)
- The key of the MCD record is the data set name (true name)
- Valid MCD information is required for Recall

Migration Records (continued)

- **MCA Record**
 - ✓ **Migration Control Data Set Alias Record**
 - Migration data set name
 - ✓ *prefix.HMIG.Tssmmhh.userqual1.userqual2.Xydd*
 - Recall intercept type
 - User data set name
- **The key of the MCA record is the DFSMSHsm generated data set name**
- **The MCA is required for tape Recycle**
- **The MCA is required for Recall of VSAM when not recalling by cluster name**

Migration Records (continued)

- MCA Record (continued)
- The MCA points back to the MCD using the DFSMSHsm generated data set name
- It is critical for this pointer to be correct when auditing tape records (TTOC). This is because the TTOC records only reference the DFSMSHsm generated data set name

Migration Records (continued)

■ MCO Record

✓ VSAM Association Record

- Cluster name
- Data component name
- Index component name
- Path name
- AIX name
- AIX data component name
- AIX index component name
- AIX path name

- The key of the MCO record is the DFSMSHsm generated name of a VSAM cluster

Migration Records (continued)

■ MCV Record

✓ Migration Control Volume Record

- VOLSER
 - Unit name
 - Device type of volume
- The key of the MCV record is:
X'04',+VOLSER, then padded to a length of 44 bytes
 - The MCV is required for Recall processing
 - There will be more than one MCV if a data set spans volumes

Migration Records (continued)

■ CDD Record

- ✓ Common Data Set Descriptor Record
 - Data set name
 - Format 1 DSCB from the VTOC
 - C '*CDD**' (eye catcher)
- ✓ Eye catcher is used for high-speed search during recall and recover processing
- ✓ CDD record must exist for this function to work

Migration Records (continued)

- **CDD Record (continued)**
- **The CDD record is the first record of every DFSMSHsm migration or backup copy**
- **The CDD record is also user by AUDIT MEDIACONTROLS**
- **If not found, AUDIT MEDIACONTROLS issues the error ERR150**

Migration Records (continued)

■ Records required for recall

✓ MCD

- All data set types

✓ MCA

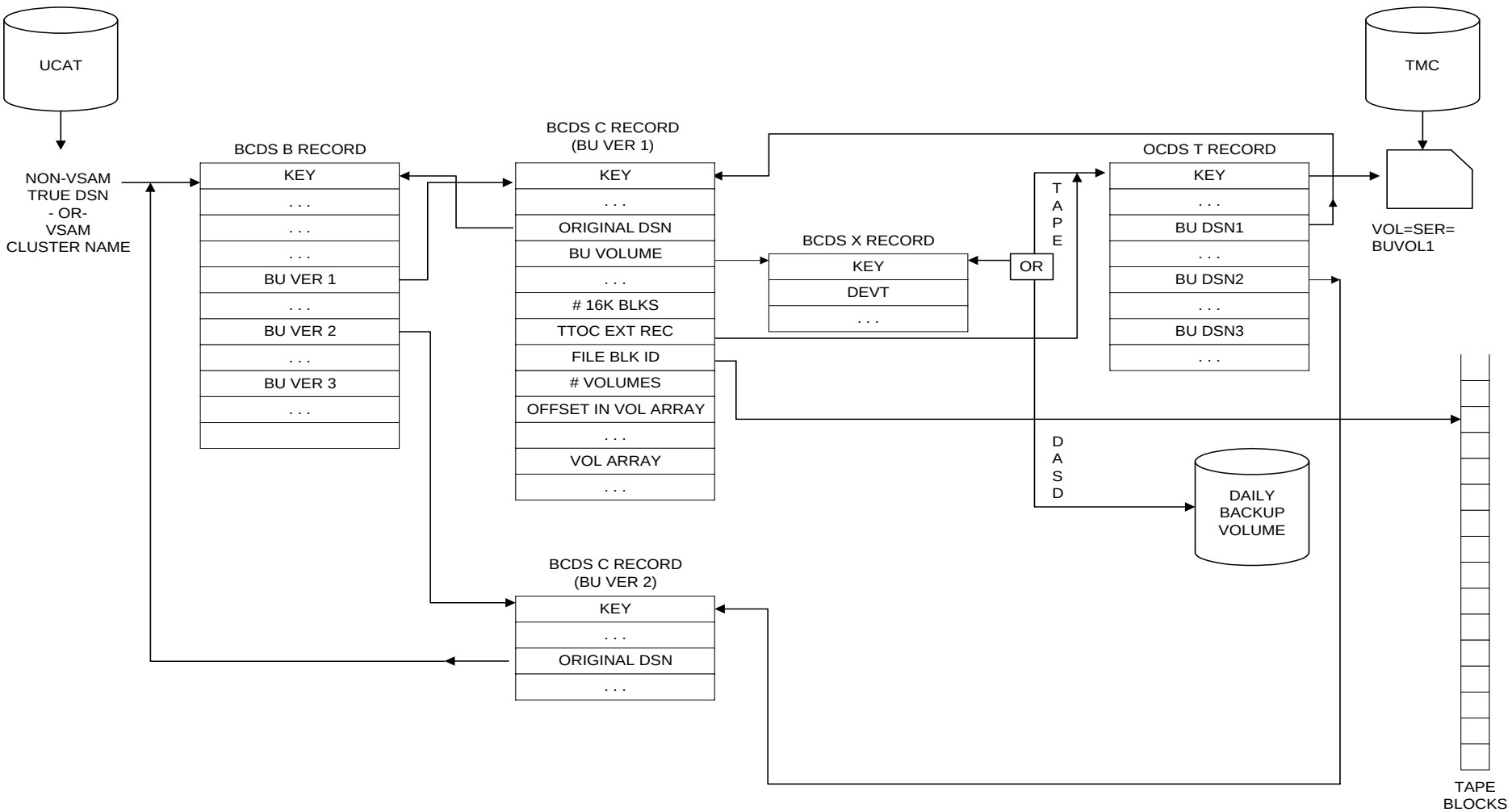
- Required to recall VSAM data sets using other than the cluster name.
- Required for RECYCLE

✓ MCV

- All data set types (tape and DASD)

✓ CDD

- All data set types



Backup Record Chains

- MCB to each MCC – DFSMSHsm generated name
- Each MCC to MCB – Data set name
- Each MCC to MCX – Daily Backup volume
 - ✓ MCX points to disk or tape volume
- MCC to TTOC (OCDS) – TTOC Extension record
- MCC to ML2 tape – File Block ID
- TTOC to MCC – DFSMSHsm generated name
- MCB to ICF Catalog – Data set name
- TTOC to Tape Management – Migration volume

Backup Records

- MCB Record
 - ✓ Backup Control Data Set Record
 - User data set name
 - Backup copy name for each version (up to 100)
 - *prefix.BACK.Tssmmhh.userqual1.userqual2.Xydd*
- The key of the MCB record is the true name of the data set

Backup Records (continued)

■ MCC Record

✓ Backup Version Record

- Backup copy name
- User data set name
- VOLSER of backup volume
- Data set type
- User volume
- Device type
- TTOC extension record
- File Block ID

- The key of the MCC record is the DFSMSHsm generated name for a particular backup version

Backup Records (continued)

- **MCT Record**

- ✓ **Backup Volume Record**

- **VOLSER**
 - **Unit name**
 - **SP, UN, or NN, and other info (2 byte flags)**

- **The key of the MCT record is:**

- X'2C'+VOLSER, then padded to a length of 44 bytes**

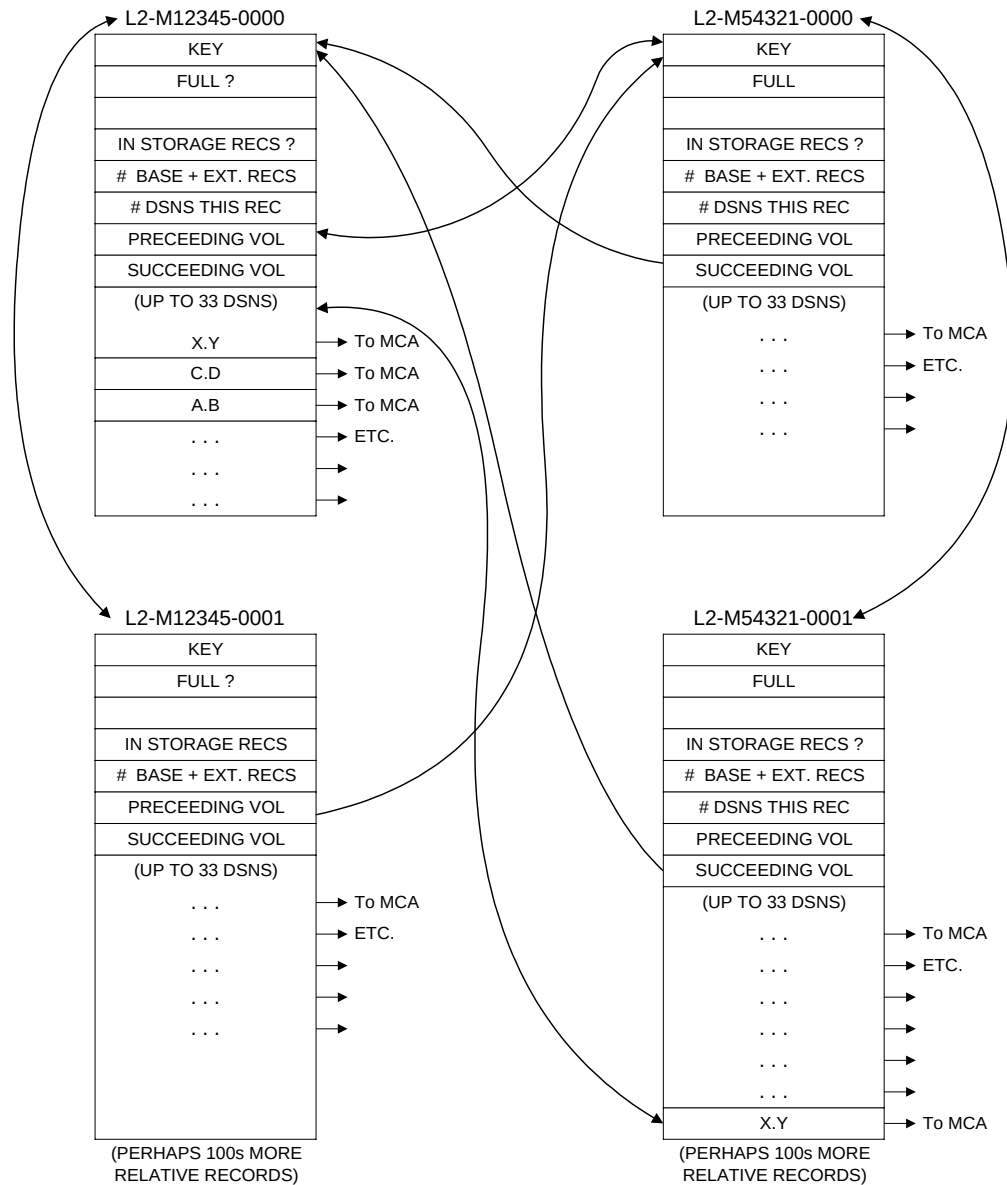
Backup Records (continued)

- **ABR Record**
 - ✓ **ABARS Backup Record**
- **Describes a specific backup version created by ABARS**
- **Simplifies the execution of an ABARS recovery command at the disaster recovery site**
 - ✓ **Contains the control file name and VOLSER**
- **ABARS backup tapes do not have TTOC records**

Backup Records (continued)

- Records required for recover
 - ✓ MCB
 - Is the driver
 - ✓ MCC
 - Backup copy name, user data set name, VOLSER, backup version, etc.
 - ✓ MCV
 - Name of primary volume
 - ✓ MCT
 - Name of backup volume
 - ✓ CDD
 - High speed search

TTOC RECORD CHAINING



Offline (Tape) Records

- **Tape Table of Contents (TTOC)**
 - ✓ Migration or backup copy data set name
- **The key of the TTOC record is:**
X'32',+SP,UN,dd,or L2,+""-",+VOLSER,+""-", then a 4-byte sequence number for a length of 44 bytes
- **TTOC records are required for recycle**
 - ✓ Not required for RECALL or RECOVER commands

Interesting Information About TTOC Records

- Arguably the most complex of the CDS records
 - ✓ DFSMShsm allows data sets to span tapes
 - Results in chained tapes
 - ✓ Most often requires several TTOC records to describe the contents of a single tape
 - ✓ Several count fields and chains between records
 - ✓ Current migration or backup tapes may contain several hundred to several thousand data sets

Auditing the DFSMShsm Control Data Sets

Reasons for Auditing Control Data Sets

- DFSMSHsm CDS recovery
 - ✓ Periodic audits reduce error conditions in recovered CDS
- Recovery of an ML1 volume
- Loss of an ML2 or backup tape
- Tapes that fail recycle
- Inability to recall or recover data
- DFSMSHsm ABENDs or a system crash
- Recovery of an SDSP

Additional Reasons for Auditing

- **Aggressive migration policies**
 - ✓ Reliance on migrated or backed-up data at home and at the disaster recovery site
- **Use of migrated data in production batch**
 - ✓ Production output files used as input to subsequent processing cycles
 - Weekly, monthly, quarterly, annual, etc.
- **Use of ABARS for application backup and recovery**

Additional Reasons for Auditing (continued)

- Reclaiming tape
- Using migration as a long-term data store
 - ✓ Historical data that is critical to the business
 - Drawings, legal records, medical information
 - ✓ Regulatory & legal requirements

IBM DFSMSHsm Command Audit and Enhanced Audit Commands:

- **AUDIT DATASET NAMES**
- **AUDIT VOLUMES**
- **AUDIT BACKUP VOLUMES**
- **AUDIT BACKUP TYPES**
- **AUDIT MASTER CATALOG**
- **AUDIT USER CATALOG**
- **AUDIT MIGRATION CONTROL DATASET**
- **AUDIT BACKUP CONTROL DATASET**

IBM DFSMSHsm Command Audit and Enhanced Audit Commands: (continued)

- **AUDIT OFFLINECONTROLDATASET**
- **AUDIT ABARSCONTROLS**
- **AUDIT DATASETCONTROLS (BACKUP)**
- **AUDIT DATASETCONTROLS (MIGRATION)**
- **AUDIT MEDIACONTROLS (SDSP)**
- **AUDIT VOLUMECONTROLS (BACKUP, MIGRATION, RECOVERABLE)**
- **AUDIT MEDIACONTROLS (VOLUMES) 'DASD and TAPE'**
- **AUDIT VOLUMES**
- **AUDIT COMMONQUEUE(RECALL)**

ISPF Command Shell

Enter TSO or Workstation commands below:

```
===> AUDIT MIGRATIONCONTROLDATASET NOFIX  
      OUTDATASET(SC.MCDS.AUDIT) REPORT(ERRORS)
```

or AUDIT ALL NOFIX

or AUDIT USERCATALOG(*catname*) NOFIX

or AUDIT BACKUPCONTROLDATASET NOFIX

or AUDIT DATASETCONTROLS(MIGRATION) NOFIX

DFSMSHsm Command Audit

-DFSMSHSM AUDIT- DFSMSHSM CONTROL DATASET - LISTING - AT 08:23:11 ON

COMMAND ENTERED:

AUDIT MIGRATIONCONTROLDATASET NOFIX OUTDATASET(SC.MCDS.AUDIT) REPORT(ERRORS)

AUDITING THE MIGRATION CONTROL DATASET

ERROR TYPE	DATASET NAME	DATASET ON VOLUME	CATA TO V	MIG TO VOL	BU TO VOL
*ERR 03	EDPTEST.LAB.SKIP.ONE	???????	-NONE	ML1001	
*ERR 03	EDPTEST.LAB.SKIP.THREE	???????	-NONE	ML1001	
*ERR 03	EDPTEST.LAB.SKIP.TWO	???????	-NONE	ML1001	
*ERR 03	EDPTEST.LAB.TEST.CREDIT	???????	-NONE	ML1001	
*ERR 03	EDPT1.LAB.ABCD2	???????	-NONE	ML1001	
*ERR 03	EDPT1.LAB.JCL	???????	-NONE	ML1001	
*ERR 03	EDPT1.LAB.VBS	???????	-NONE	ML1001	
*ERR 03	EH.EIGHTY.BYTES	???????	-NONE	ML1001	
- END OF -	CATALOG - LISTING -				

To Fix or Not to Fix

- **FIX or NOFIX is an optional parameter**
- **Default is NOFIX**
 - ✓ NOFIX only reports error conditions
 - ✓ User must code FIXCDS commands to resolve
- **FIX uses the data set's ICF catalog status in determining repair action**
 - ✓ For instance, FIX deletes the CDS records of uncataloged data sets
 - ✓ This can be an issue if you have ICF catalog problems or if data sets were inadvertently uncataloged

Limiting the scope of the audit

- Limit to Control Data Set (MCDS, BCDS, OCDS)
- Limit to specific volumes (or list of volumes)
- Limit to specific ICF catalogs
- Limit to specific data sets (or list of data sets)
- Limit to groups of like-named data sets (LEVEL parameter)

Two Different Audits

■ AUDIT ABARSCONTROLS

- ✓ Audits the ABR record in the BCDS
- ✓ Reports missing ABR records for specified aggregate
- ✓ Reports missing or uncataloged control, data, or instruction files
- ✓ No fixes are performed

■ AUDIT COMMONQUEUE(RECALL)

- ✓ Used to audit Common Recall Queue (CRQ)
- ✓ Messages ARC1506E and ARC1187E may indicate CRQ corruption
- ✓ Specifying FIX may result in less reported errors than when NOFIX is specified, since one fix may correct multiple errors

AUDITMEDIACONTROLS Audits Tape

- During AUDITMEDIACONTROLS executions
 - ✓ DFSMSHsm reads the entire contents of the backup or migration tape
 - Verifies that most fields in the TTOC records are accurate
- Tapes in Failed Create status
 - ✓ Caused by incorrect TTOC records
 - Often due to missing records
 - How does this happen?

Migration Tapes in Failed Create Status

- **During DFSMSHsm Migration**
 - ✓ **DFSMSHsm attempts to reduce I/O to the OCDS**
 - Accumulates in storage up to 33 changes to any TTOC record or six hours' worth of changes, whichever occurs first
 - ✓ **Sets a flag indicating that in-storage records exist and it turns that flag off after it has updated the TTOC**

Migration Tapes in Failed Create Status (continued)

- If you cancel DFSMShsm, or it is otherwise halted, without writing these records out to the tape
 - ✓ Migration tape is put in failed create status (ARC0378I)
 - ✓ Tape is not available for recycle
 - ✓ You must execute AUDITMEDIACONTROLS

Creating a List of Tapes to Audit

- Failed recycle

HSEND LIST TTOC SELECT(BOTH FAILEDRECYCLE)

- Failed create

HSEND LIST TTOC SELECT(BOTH FAILEDCREATE)

DFSMSHsm Resources

- **Additional resources are required when executing AUDITMEDIACONTROLS**

- ✓ **Time**

- Several hours, possibly several days, to audit even one tape

- ✓ **DFSMSHsm**

- CDS backup is held until tape audit is complete
- Audit cannot be restarted from stopping point
- Audit executes as a low priority task

- ✓ **Hardware**

- Tape drives held and unusable during audit

Regular DFSMSHsm Audits Are Necessary

- How frequently is dependent upon your use of DFSMSHsm
- Regular auditing
 - ✓ Ensures data is accessible
 - Service is not interrupted
 - ✓ CDS recovery is faster
 - Results are more reliable
 - ✓ DR recovery
 - Synchronization of ICF catalogs with Control Data Sets is fast and accurate

- ***DFSMSHsm Storage Administration Guide (SC35-0421)***
- ***DFSMSHsm Storage Administration Reference (SC35-0422)***
- ***DFSMSHsm Diagnosis Reference (LY35-0111)***
- ***DFSMSHsm Audit and Mainstar: FastAudit/390 (SG24-6425)***

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